SYMPHONY-COORD

#1 Task & Plan-K

Incoming task

TASK

q

Plan builder

PLAN

K candidate plans

PLAN $t_{k,1} \rightarrow \dots \rightarrow t_{k,L_k}$

#2 Capability-Guided Screening (CGS)

Current subtask

$t_{k,m}$ + memory H

Available pool $\mathcal{A}_t$

CGS: Top-L

match | prior | reliability

Shortlist C_t

#3 Contextual Bandit Routing (CBR) & CoT-P

delayed feedback

CBR: LinUCB

match, load, latency
prior, availability

estimate + uncertainty

shared state A, b

Retained action record

a_t + selected context

P independent routed executions

run 1 | run 2 | ... | run P

Normalize + majority vote

Chain complete: plan output y_k

#4 Aggregate & Learn

Completed plan outputs

y_1 y_2 ... y_K

Weighted cross-plan vote

w_k: mean CGS score

Final answer

y^*

Post-vote reward

r

+ validity | + agreement

optional correctness | − latency

Update A, b when resolved